

Analyzing Climate and Spectral Drivers of Forest Loss in the Acadian Region using Machine Learning

Benjamin Pace
bfp2113

May 11, 2025

Abstract

Understanding drivers of forest change is critical under a changing climate. This paper documents the development and execution of a machine learning pipeline to analyze non-fire related forest loss in a Northern Maine study area (target years 2001-2021). The project initially leveraged prior group exploration of spectral tree species classification using Sentinel-2 imagery and GBIF occurrence data. These attempts, employing both Random Forest and Convolutional Neural Network (CNN) models, faced challenges including data sparsity, high spectral similarity among species, and critical technical data extraction failures resulting in low preliminary model performance (e.g., 29% RF accuracy for species ID; CNN performance was similarly constrained). This necessitated a strategic pivot to the current methodology, which predicts aggregate annual forest loss (Hansen GFC) using a Random Forest classifier with time-lagged Landsat C2 L2 composites, TerraClimate data, and SRTM topography. This paper details data acquisition (GEE), pre-processing (including merging GEE-tiled Landsat exports and rigorous alignment verification), pixel sampling strategies (with downsampling for this report due to NaN issues and time constraints), feature engineering with NaN imputation, and a temporal evaluation framework. Initial final product results (target years 2001-2002, 100k downsampled points) yielded an ROC AUC of 0.56, highlighting data quality sensitivities but establishing pipeline functionality. The work emphasizes the iterative research process, problem-solving for data challenges, and lays the groundwork for full-scale analysis. The final code is available in the project repository available at <https://github.com/9pace/mlcs>.

1 Introduction

Forest ecosystems worldwide are experiencing significant changes due to anthropogenic activities and a shifting climate [1]. The Acadian forest region (Figure 1), particularly Northern Maine, faces pressures from climate stress, pests, and logging, distinct from large-scale wildfire events common in other geographies. Understanding the drivers of this non-fire related forest loss is vital for effective forest management, conservation planning, and predicting future landscape dynamics under continued environmental change. This project aimed to develop a machine learning (ML) pipeline for this purpose, focusing on an interpretable model to identify key predictive factors.

Initially, the project's ambition was to track species-specific responses to environmental pressures using advanced ML techniques. However, practical data limitations encountered during early exploration, a common experience in applied ecological remote sensing, necessitated a methodological refocus. This paper details this evolution. Section 2 describes the initial species classification attempts and the challenges that prompted a pivot. Section 3 outlines the current methodology for analyzing aggregate forest loss drivers. Section 4 presents initial findings from a final product execution of this revised pipeline, designed to test core functionality. Section 5 briefly touches on the project management and iterative development process. Finally, Section 6 discusses the broader implications of these results, limitations of the current work, and potential avenues for future research. The final analysis code, implemented in a Jupyter

Notebook (01_forest_loss_analysis.ipynb), is located in the notebooks/ directory of the project's GitHub repository available at <https://github.com/9pace/mlcs>.

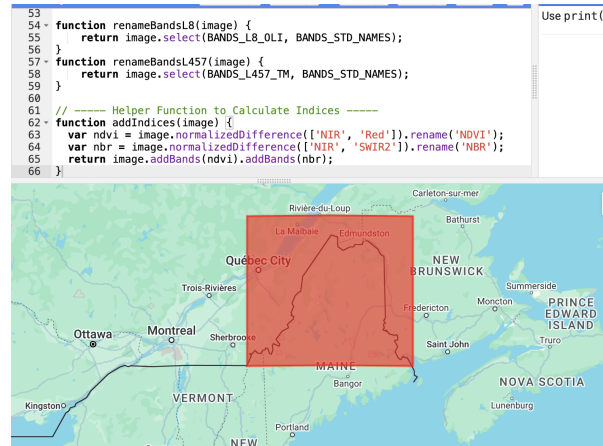


Figure 1: Study Area: Northern Maine within the Acadian Forest Region, showing the Google Earth Engine export boundary used for data acquisition.

2 Initial Exploration & Challenges: Species Classification

The project's first phase, building upon prior collaborative work, focused on identifying individual, climate-sensitive tree species (e.g., *Acer rubrum*, *Abies balsamea*) using Sentinel-2 satellite imagery and species occurrence data sourced from the Global Biodiversity Information Facility (GBIF) [2]. Both Random Forest (RF) [3] and 1D Convolutional Neural Network (CNN) models were preliminarily tested for this classification task, drawing on literature that has explored such methods for vegetation analysis [4–6].

Despite the potential of this approach, several significant challenges emerged, underscoring the complexities of species-level remote sensing with standard multispectral data:

- **Species Data Sparsity and Quality:** GBIF [2] data, while globally extensive, often resulted in sparse or spatially clustered occurrence points for specific target species within our defined study area (illustrated conceptually in Figure 2a). This limited the ability to obtain a large, representative, and spatially balanced training dataset crucial for supervised classification, a common issue when relying on opportunistic occurrence records.
- **High Spectral Similarity Among Species:** Analysis of extracted Sentinel-2 spectral signatures revealed a critical challenge: high similarity between the target tree species across the available bands (Figure 2b). For example, the average reflectance values for species such as *Pinus strobus*, *Abies balsamea*, *Acer rubrum*, *Tsuga canadensis*, *Betula papyrifera*, and *Picea abies* showed overlapping profiles, particularly in the visible and some red-edge regions. While subtle mean differences existed, the variance within species and the overlap between species made their distinct spectral separation extremely difficult using these standard multispectral bands alone. This inherent lack of strong spectral separability is a known fundamental challenge in remote sensing of mixed forests, especially when not employing hyperspectral data or very high-resolution imagery capable of capturing fine textural or structural differences [6].
- **Technical Data Extraction Failures:** A critical technical hurdle was the reliable extraction of spectral data corresponding to the GBIF points. Scripts designed to sample pixel values from Sentinel-2 tiles at these point locations frequently failed to match coordinates precisely or reported a high percentage of points as falling outside image boundaries. For instance, one key script iteration attempting to extract signatures from approximately 7,700 GBIF points for Autumn 2018 Sentinel-2 imagery reported that 100% of points were deemed outside image boundaries, effectively halting feature generation for that

dataset. This highlighted complexities in managing coordinate systems, raster tiling, and precise point-in-polygon operations with large satellite imagery archives.

- **Low Preliminary Model Performance:** Consequently, preliminary RF models trained on the limited and potentially noisy spectral data that could be extracted yielded poor classification performance, with test accuracies hovering around 29% and high inter-class confusion (Figure 2c). The confusion matrix, for instance, frequently showed misclassifications between spectrally similar deciduous species (e.g., Maples and Birches if they were grouped) or between different coniferous types (e.g., Pines and Firs). The experimental CNN models, despite their capacity for learning complex hierarchical features, did not yield significantly better results. Their performance was similarly hampered by the fundamental limitations of the input data: the spectral non-distinctiveness shown in Figure 2b meant there were insufficient unique spectral patterns for the CNN to effectively learn class boundaries. The similar, unsatisfactory outcomes from both classical ML (Random Forest) and a basic deep learning approach (CNN) strongly suggested the bottleneck was primarily data-driven (i.e., the spectral information itself was not sufficiently separable for the task) rather than a limitation of the chosen model architectures for this specific problem.

The combination of these issues made the species-specific classification approach infeasible to pursue effectively within the project's timeframe and available resources. This led to a strategic pivot towards analyzing aggregate forest loss, a phenomenon for which more robust and spatially complete data were available.

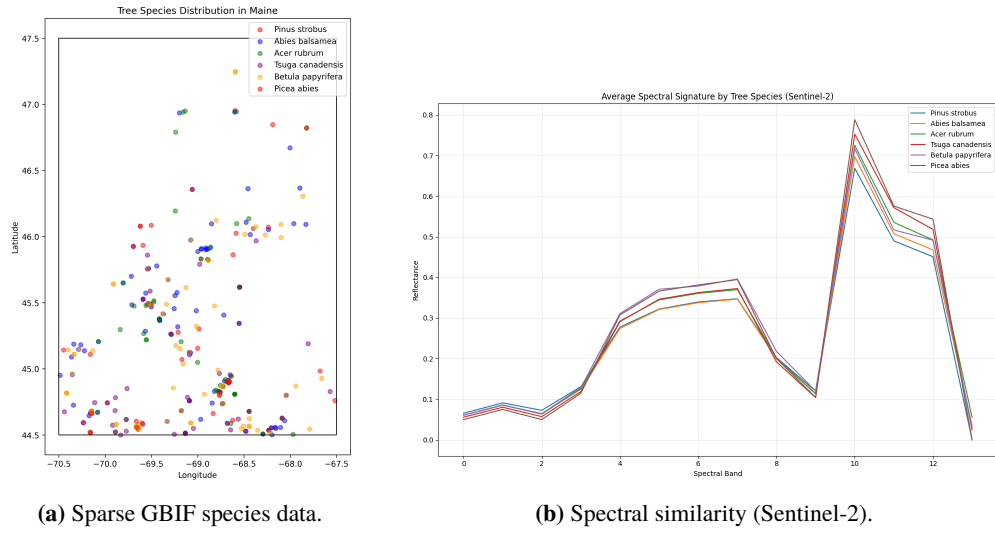


Figure 2: Illustrations from the initial species classification exploration phase: (a) Example of sparse GBIF species point data. (b) Average spectral signatures from Sentinel-2 showing high similarity among species, making them hard to distinguish. (c) Confusion matrix from an early RF model for species classification indicating low accuracy and high inter-class confusion.

3 Revised Methodology: Forest Loss Driver Analysis

The revised objective is to predict non-fire forest loss in a target year ‘t’ (2001-2021) in Northern Maine using a Random Forest model based on conditions in ‘t-1’ and ‘t’. The overall workflow is depicted in Figure 3.

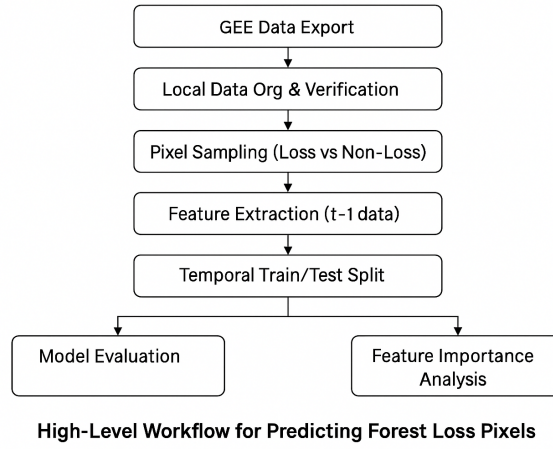


Figure 3: High-level workflow for the forest loss prediction pipeline, from GEE data export to model evaluation and feature importance analysis.

3.1 Data Sources and Preparation

Key datasets include (Figure 4):

- **Landsat & Hansen GFC:** Landsat C2 L2 annual median composites (2000-2021, 30m, GEE [7]) provided six spectral bands (Blue, Green, Red, NIR, SWIR1, SWIR2) and derived NDVI/NBR. Hansen GFC v1.9 [8] provided `lossyear` (target variable indicating year of loss) and `treecover2000` (for masking stable forest areas).
- **TerraClimate & SRTM:** Annual TerraClimate [9] mean temperature and total precipitation (~4km resampled to 30m via GEE) and SRTM [10] DEM/Slope (30m) were used as environmental predictors.

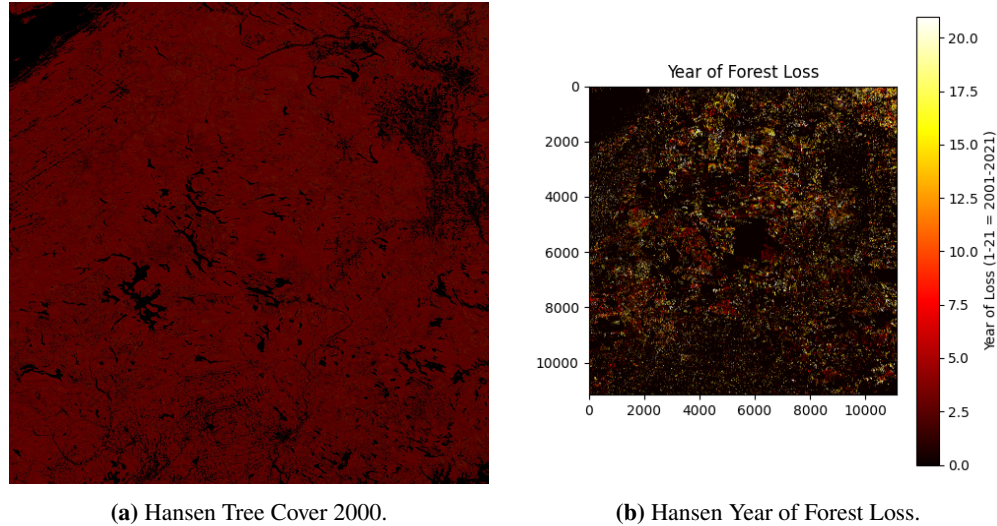


Figure 4: Example Hansen GFC data products: (a) Baseline tree cover in 2000 used for masking stable forest. (b) Year of forest loss (target variable).

3.1.1 Data Alignment and Tiling Solution

Ensuring spatial alignment of all ~70 rasters was critical, verified using a Python function with ‘rasterio’ [11]. GEE-exported Landsat composites were often tiled; the command-line tool `gdal_merge.py` (from GDAL [12]) was essential for mosaicking these tiles locally into single annual files.

3.2 Pixel Sampling Strategy

For each target year 't': Loss pixels were identified. Non-loss control pixels were randomly sampled (ratio 1:1 for final product run) from stable forested areas. For the final product results, the total sampled points across 2001-2002 target years were aggressively downsampled from ~2.84 million to ~100,000 due to extensive NaN values and time constraints.

3.3 Feature Engineering and NaN Handling

Features extracted per pixel (target year 't') using 'rasterio' [11] included: Landsat (year 't-1'), Climate (year 't-1'), Climate (year 't'), and Static (DEM, slope). Significant NaN values encountered in the final product run (e.g., temp_2000 often all NaN for the subset) were handled via mean imputation (zero for all-NaN columns).

3.4 Model and Evaluation

A Random Forest Classifier (scikit-learn [3]) with `class_weight='balanced'`, `n_estimators=10`, and `max_depth=5` was used. Data was split temporally (Figure 5). For the MVP, training used data linked to target year 2001, and testing used data from target year 2002. Performance was assessed using overall accuracy, precision, recall, F1-score, confusion matrix, and ROC AUC.

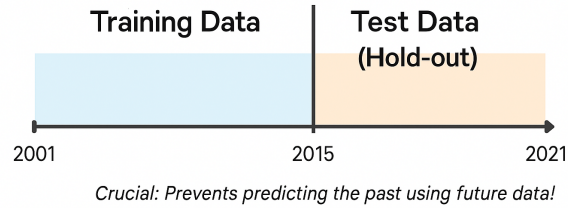


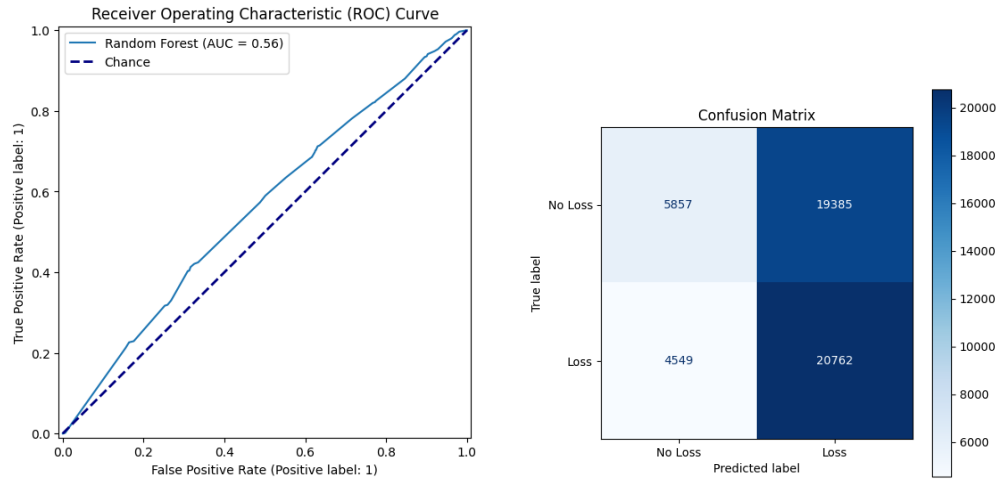
Figure 5: Idealized temporal train/test split strategy.

4 Final product Results & Preliminary Findings

The analysis pipeline was executed with a reduced scope (target years 2001-2002, ~100,000 downsampled points, simplified RF parameters) to demonstrate core functionality.

4.1 Model Performance

The RF model yielded an overall accuracy of ~53% on the test set. For the 'Loss' class, Precision=0.52, Recall=0.82, F1-Score=0.63. The ROC AUC Score was 0.56 (Figure 6a). The confusion matrix is in Figure 6b. This performance, slightly better than random, reflects the Final product's limitations.



(a) ROC Curve (Final product Test Set, AUC=0.56). (b) Confusion Matrix (final product Test Set).

Figure 6: Performance of the Random Forest model on the final product test set (target year 2002).

4.2 Feature Importances

Despite the modest overall performance of the MVP model, the feature importance analysis (Figure 7) from the Random Forest run provides some initial, albeit tentative, insights into which variables the model found most useful for its limited discrimination, given the heavily downsampled and imputed dataset.

The top-ranked features were predominantly derived from Landsat imagery of the preceding year ($t-1$, which was year 2000 for this MVP run predicting 2001/2002 loss). Specifically, `lsat_SWIR1_2000` (Shortwave Infrared 1) and `lsat_SWIR2_2000` (Shortwave Infrared 2) emerged with the highest Gini importance values. SWIR bands are well-known for their sensitivity to vegetation water content, canopy structure, and the presence of non-photosynthetic vegetation (e.g., dry matter, woody material, senescing leaves) [13]. Elevated importance for these bands might suggest that pre-loss stress conditions, possibly related to drought stress or initial stages of dieback leading to changes in canopy moisture or structure, were detectable signals the model keyed in on.

Following the SWIR bands, `lsat_NIR_2000` (Near-Infrared) also showed relatively high importance. NIR reflectance is strongly correlated with healthy, dense vegetation due to light scattering in the leaf mesophyll. A decrease in NIR reflectance in year $t-1$ could indicate reduced leaf area, chlorophyll degradation, or other stressors, making it a plausible precursor signal for subsequent loss.

Interestingly, `precip_2000` (total precipitation from the year 2000, serving as a $t-1$ or $t-2$ predictor depending on the target loss year in the MVP) also ranked among the top predictors. This suggests that antecedent precipitation conditions (e.g., a drought year or perhaps an unusually wet year leading to other stresses like waterlogging or fungal outbreaks) might play a significant role in predisposing forest areas to loss in subsequent years.

The Landsat Green and Red bands from year 2000, as well as DEM (elevation) and slope, showed moderate importance. Broader vegetation indices like `lsat_NDVI_2000` had lower importance in this specific MVP run than some individual spectral bands. This could be an artifact of the particular data subset, the model's simplified parameters, or perhaps that the individual bands, particularly in the SWIR and NIR regions, captured more specific stress signals than the normalized difference of NDVI for this limited dataset.

It is crucial to note that the negligible importance of several climate features, such as `temp_2000`, `precip_2001` (for some target years), `temp_2002`, and `precip_2002`, is highly likely an artifact of the MVP data processing. As documented, these columns contained a high proportion of, or

were entirely, NaN values for the processed subset and were subsequently imputed (often with zero). This imputation would effectively neutralize their predictive power in the Random Forest model, leading to artificially low importance scores. Therefore, the current feature importance ranking should be interpreted with significant caution, primarily as an indicator of what the model could leverage from the available, non-imputed (or less imputed) data within the MVP's severe constraints. A full analysis with more robust data handling is required for definitive conclusions on feature importance.

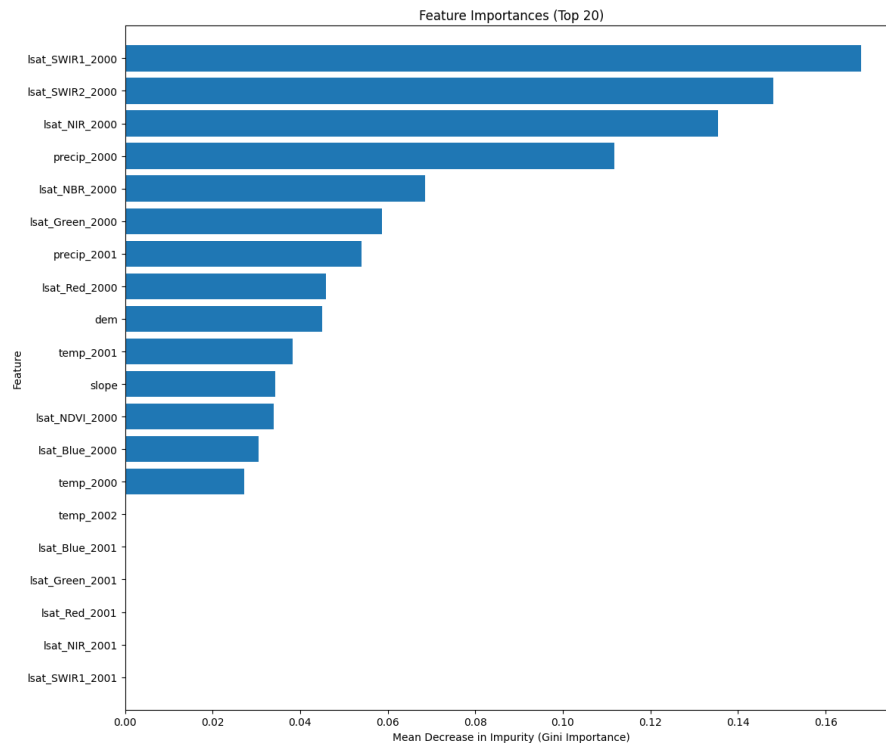


Figure 7: Top 20 Feature Importances from the MVP Random Forest model. Note: The low importance of some year-specific climate variables (e.g., `temp_2000`, `temp_2002`) is likely influenced by their imputation from all-NaN states for this particular data subset.

5 Project Management and Iterative Development

This project followed an iterative solo development process from April 16th, 2025. Initial efforts focused on scope refinement and data acquisition (GEE scripts finalized April 18th-20th). The ML pipeline was developed April 21st-24th. Significant time (April 26th-May 3rd) was spent addressing NaN data quality issues, leading to the final product strategy with aggressive downsampling. Local Git commits preserved original author dates; history was rewritten before final GitHub submission (May 9th-11th) to manage large file sizes using `git filter-repo`, resulting in a batched appearance of commits on the remote platform.

6 Discussion & Future Work

Final product results (ROC AUC 0.56) are modest, primarily due to aggressive downsampling, a limited target scope, and NaN imputation challenges. However, pipeline functionality is verified. Identified feature importances (SWIR, NIR, antecedent precipitation) are ecologically plausible but need validation with the full dataset.

Limitations include coarse TerraClimate resolution [9] and aggregated Hansen GFC loss types [8]. Future work requires running on the full, non-downsampled dataset with robust NaN handling, hyperparameter tuning, and exploring other models or spatial features.

7 Conclusion

This project successfully pivoted from a challenging species classification task to develop a comprehensive pipeline for analyzing aggregate forest loss drivers. The final product execution, despite data limitations, validated the pipeline’s functionality. The established framework provides a solid foundation for future full-scale analysis to generate significant insights into non-fire forest loss in the Acadian region. The final, executable code demonstrating this final product pipeline is in `01_forest_loss_analysis.ipynb` within the project repository available at <https://github.com/9pace/mlcs>.

References

- [1] Rupert Seidl et al. “Forest disturbances under climate change”. In: *Nature Climate Change* 7.6 (2017), pp. 395–402. doi: [10.1038/nclimate3303](https://doi.org/10.1038/nclimate3303).
- [2] GBIF.org. *GBIF Home Page*. 2024. URL: <https://www.gbif.org> (visited on 05/06/2024).
- [3] F. Pedregosa et al. “Scikit-learn: Machine Learning in Python”. In: *Journal of Machine Learning Research* 12 (2011), pp. 2825–2830.
- [4] Imran Md Jelas et al. “Deforestation detection using deep learning-based semantic segmentation techniques: a systematic review”. In: *Frontiers in Forests and Global Change* 7 (Feb. 2024). Section: Forest Management, p. 1300060. doi: [10.3389/ffgc.2024.1300060](https://doi.org/10.3389/ffgc.2024.1300060).
- [5] R. Shanmuga Priya Rajendran and K. Vani K. “Vegetation Change Detection and Recovery Assessment on Post-fire Satellite Imagery using Deep Learning”. In: (Jan. 2024). Research Square Preprint, available at <https://doi.org/10.21203/rs.3.rs-3890182/v1>. Licensed CC BY 4.0. doi: [10.21203/rs.3.rs-3890182/v1](https://doi.org/10.21203/rs.3.rs-3890182/v1).
- [6] Somayeh Nezami et al. “Tree Species Classification of Drone Hyperspectral and RGB Imagery with Deep Learning Convolutional Neural Networks”. In: (Feb. 2020). Preprint available at <https://www.preprints.org/manuscript/202002.0334/v1>. doi: [10.20944/preprints202002.0334.v1](https://doi.org/10.20944/preprints202002.0334.v1).
- [7] Noel Gorelick et al. “Google Earth Engine: Planetary-scale geospatial analysis for everyone”. In: *Remote Sensing of Environment* 202 (2017), pp. 18–27. doi: [10.1016/j.rse.2017.06.031](https://doi.org/10.1016/j.rse.2017.06.031).
- [8] M. C. Hansen et al. “High-Resolution Global Maps of 21st-Century Forest Cover Change”. In: *Science* 342.6160 (2013), pp. 850–853. doi: [10.1126/science.1244693](https://doi.org/10.1126/science.1244693).
- [9] John T. Abatzoglou et al. “TerraClimate, a high-resolution global dataset of monthly climate and climatic water balance from 1958–2015”. In: *Scientific Data* 5 (2018), p. 170191. doi: [10.1038/sdata.2017.191](https://doi.org/10.1038/sdata.2017.191).
- [10] NASA JPL. *Shuttle Radar Topography Mission (SRTM)*. 2015. URL: <https://www.jpl.nasa.gov/missions/shuttle-radar-topography-mission-srtm/> (visited on 05/06/2024).
- [11] Sean Gillies et al. *Rasterio: Geospatial raster CAPI for Python*. Version 1.3.10 used in this project. 2013–. URL: <https://github.com/rasterio/rasterio>.
- [12] GDAL/OGR contributors. *GDAL/OGR Geospatial Data Abstraction software Library*. Version 3.8.5 used in this project. Open Source Geospatial Foundation. 2024. URL: <https://gdal.org>.
- [13] C. J. Tucker. “Remote sensing of leaf water content in the near infrared”. In: *Remote Sensing of Environment* 10.1 (1980), pp. 23–32. doi: [10.1016/0034-4257\(80\)90096-6](https://doi.org/10.1016/0034-4257(80)90096-6).